An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

1. Executive summary

Floods displace millions of people in India every year. The hard part of a rescue is not reaching survivors; it is finding them.

We want to build three small aircraft, about a metre across and 6.4 kg each, that search together. One person sends them out over flooded ground. They fly themselves, spot people from the air, work out where those people are to within about a metre, and drop a relief kit to each. Nothing depends on a phone network or the internet.

The design and the software are done, and we have flown the whole mission in simulation with three aircraft cooperating. Nothing has flown for real yet, so every figure here is calculated rather than measured.

We would like to begin with INR 3,733: enough to test whether one motor lifts what the design needs. After that the money comes in 29 small steps, each released only once the step before it has produced a result we can show you. Nothing over INR 26,000 is spent until that first thrust measurement exists. If every step is approved the programme comes to INR 6,85,532.

2. The problem

Boats are slow, dangerous for the crew, and cannot see onto rooftops. Helicopters are few and are needed for evacuation. Ordinary drones need a pilot each, rely on infrastructure the flood has washed away, and cannot help anyone once they have found them.

Finding people turns out to be the harder half. Someone in floodwater shows only their head and shoulders — about 40 cm of them is visible from above. Detection software works on a small picture of fixed size, so the usual approach shrinks the photograph, and the person shrinks with it until there is too little left to recognise. Using a smaller, cheaper model makes that worse, not better.

There is also no public set of labelled flood photographs to learn from. Thermal cameras do not rescue us either: someone sitting in water ends up close to the temperature of the water.

3. Our solution

The operator draws a search area of any shape. The software splits it three ways and plans a route over each share. We have tried this on thirteen awkward shapes and no route ever wandered outside the area. After that the aircraft fly themselves, and our tests confirm nothing is commanded from the ground once a mission has begun. Each aircraft spots people and works out where they are on its own, carries four relief kits, descends over a survivor, holds still, and drops one. If the radio fails, the battery runs low, or the operator hits abort, all three end the same way: come home and land.

The trick with the camera is not to shrink the picture. We cut each frame into tiles and search them at full size, and we fly lower. Between them those two changes make a person in the water nine times bigger in the image, which is what lifts them clear of the range where this kind of software stops working. Neither change is enough on its own. It costs four times as much computing, which the accelerator we have chosen can supply, and the mission takes no longer, because flying lower saves more climbing than the extra passes cost.

There is one thing we have not solved. We plan to look at each patch of ground twice a second, which the accelerator manages comfortably. To be really confident about a sighting we would want three looks a second, and that is half as much work again — more than we can count on. There may be a way round it: most of what the camera sees is plainly open water, so a quick, cheap check could throw those patches away and leave the expensive search to the rest. The danger is that if the quick check throws away a patch with a person in it, nothing tells us. No error, no warning, just someone never found. We will measure how often that happens against public data before we rely on it.

4. Deliverables

1. Three aircraft operating as a coordinated group under one operator.
2. A ground station that shows what is happening and, by design, cannot fly the aircraft.
3. The autonomy software, with an automatic test suite behind it.
4. Measured detection accuracy and a verdict on the two-stage filter.
5. A publication: tiling, aerial person detection and low-power onboard inference have not been combined for flood conditions.
6. Source code and results, published in either outcome.

5. Benefits

- Ten hectares searched in under eight minutes, and what comes back is a set of coordinates a rescue team can act on, not a video feed somebody has to sit and watch.
- The whole thing costs less than three hours of helicopter charter at Indian rates, travels by road, and is run by two people.
- A thrust stand, a ground station, a tested codebase and students who have built all of it — and all of that stays in the department afterwards.
- Every part but one comes from an Indian supplier, so nothing sits at customs and anything broken in testing is replaced within days.

6. Resources needed

Funding is requested in 29 small steps rather than as one grant. Each step buys one named set of parts and is released only once the previous step has produced a stated result. If a step fails, the programme stops there and nothing beyond it has been spent.

The order follows engineering dependency: one motor is measured before the other eleven are bought, and the first aircraft flies a complete mission before the second is begun.

Phase schedule

The third column is the amount released. The fourth is what must already be true before it is released.

#	Objective	INR	Released only after
1	Establish thrust and mass measurement capability	3,733	Approval
2	Measure one motor, propeller and speed controller on the stand	13,263	Instruments commissioned
3	Aircraft 1 — autopilot and control link	27,323	**Thrust verified**
4	Aircraft 1 — onboard computer and AI accelerator	41,694	Autopilot bench-tested
5	Aircraft 1 — centimetre positioning: RTK rover and ground base	67,850	Compute stack operating
6	Aircraft 1 — camera and the three radios	19,614	Position fix acquired

#	Objective	INR	Released only after
7	Aircraft 1 — airframe fabrication	18,886	Avionics integrated
8	Aircraft 1 — battery pack and power distribution	35,527	Airframe fabricated
9	Aircraft 1 — release servos and wiring	3,075	Pack bench-discharged
10	Aircraft 1 — propulsion completed	25,661	Drive train installed
11	Charging, pack instrumentation and manual override	37,865	Aircraft assembled and weighed
12	Field consumables and assembly materials	7,934	Charging verified
13	Aircraft 2 — autopilot and control link	27,323	**Aircraft 1 flew a full mission**
14	Aircraft 2 — onboard computer and AI accelerator	41,694	Autopilot bench-tested
15	Aircraft 2 — satellite positioning and compass	7,980	Compute stack operating
16	Aircraft 2 — camera and the three radios	19,614	Position fix acquired
17	Aircraft 2 — airframe fabrication	17,795	Avionics integrated
18	Aircraft 2 — battery pack and power distribution	26,028	Airframe fabricated
19	Aircraft 2 — speed controllers, release servos, wiring	8,871	Pack bench-discharged
20	Aircraft 2 — propulsion completed	31,070	Drive train installed
21	Aircraft 3 — autopilot and control link	27,323	**Two aircraft, separation verified**
22	Aircraft 3 — onboard computer and AI accelerator	41,694	Autopilot bench-tested
23	Aircraft 3 — satellite positioning and compass	7,980	Compute stack operating
24	Aircraft 3 — camera and the three radios	19,614	Position fix acquired
25	Aircraft 3 — airframe fabrication	17,800	Avionics integrated
26	Aircraft 3 — battery pack and power distribution	26,028	Airframe fabricated
27	Aircraft 3 — speed controllers, release servos, wiring	8,871	Pack bench-discharged
28	Aircraft 3 — propulsion completed	31,070	Drive train installed
29	Ground station: three video feeds, data link, base mount	22,351	Third aircraft flying
	Total	**6,85,532**	

Phase 3 commits the first significant expenditure and is released only on a measured thrust figure. Phase 13 begins the second aircraft only after the first has flown a complete mission, so the largest blocks of expenditure fund replication of a demonstrated configuration rather than a projection.

Component selection

The schedule gives the amount *released* per phase. The tables below give the parts each phase *buys*. A released amount is the parts cost plus tax where tax is still owed, plus 15 % contingency. Across the programme that is INR 5,81,034 of parts against INR 6,85,532 released.

Every line below is a current listing from a named Indian supplier, priced and linked. An earlier version said the opposite — a handful firm, the rest estimates. The schedule and the tables are generated from that same list, so they cannot drift apart.

Four changes have between them *reduced* the ask by about INR 67,000, after adding back two components a review found missing. The onboard computer was budgeted at INR 8,000 against a real INR 19,999. The certified scale has left the ask entirely, because the department owns one. The earlier version added customs duty and GST to every line, which was right for estimates but wrong for Indian retail prices, where the tax is already paid. And only one aircraft now carries the costly centimetre-accurate receiver: all three are the same machine, so measuring one against surveyed ground tells us how well the design works, and the other two fly a two-metre receiver against a five-metre requirement. That last change alone saved INR 35,600.

Terms used below. *RTK*: satellite positioning corrected against a second receiver on a surveyed point, giving centimetre rather than metre accuracy. *6S3P*: eighteen cells, six in series for voltage and three in parallel for capacity. *kgf*: thrust written as the weight it would lift. *TOPS*: trillion operations per second, the accelerator's throughput. *ESC*: the electronic speed controller between battery and motor. *BMS*: the board that balances and protects the cells — there is none in this build, and the note after the tables says why.

Measurement instruments — phases 1–2 INR 3,246 of parts

Item	Model	INR, all 3	Why this part, and from whom
Load-cell amplifier	7Semi HX711 24-bit ADC	104	Turns the load cell's signal into a reading we can log. *Robu*
Load cell	REES52 20 kg load cell + HX711 breakout	854	The motors ship without a thrust curve, so we measure it ourselves. Ready-made stands cost around ten times this. *Amazon India*
Thrust-stand mast	SURYAKUSH metal 2.1 m tripod	373	Holds the motor clear of the bench, so its own draught does not flatter the reading. *Flipkart*
Fasteners	Hex socket-head assortment M3/M4/M5, SS304	1,140	A carbon airframe is bolted, not glued. *OnlyScrews*
Threadlocker	Loctite 243, 50 ml	664	Stops bolts working loose under motor vibration. Medium strength, so a joint can still be undone. *Amazon India*
Throttle source	Servo tester / ESC consistency tester, PPM	111	A motor on a stand needs something to command it. The safety-pilot transmitter is not bought until phase 11, long after this test. *Robokits*

Per aircraft, avionics and sensing — phases 3–6, repeated at 13–16 and 21–24 INR 85,775 per aircraft, INR 3,21,203 in all

Item	Model	INR, all 3	Why this part, and from whom
Flight controller	Holybro Pixhawk 6C Mini, 3 off	66,147	The autopilot: flies the aircraft and runs its safety behaviours. Previously a quotation, now ordinary retail. *Robu*
FC vibration mount	Glass-fibre anti-vibration shock absorber, 3 off	483	Isolates the autopilot from propeller vibration, which otherwise confuses its sensors. *Robu*
Autopilot log card	SanDisk 32 GB microSDHC, class 10, 3 off	4,647	The autopilot ships without one and records nothing without it. This is a different slot on a different board from the computer's card above. *Robocraze*
Companion computer	Raspberry Pi 5, 8 GB, 3 off	59,997	The computer the autonomy runs on. The accelerator and the camera both plug into it, so neither works without it. *Robu*
AI accelerator	Raspberry Pi AI HAT+, 26 TOPS, 3 off	35,247	Runs the person-detection model fast enough to search at flying speed. *Robu*
Compute cooling	Official Raspberry Pi 5 active cooler, 3 off	1,557	The hottest moment is the setup on the ground, before the propellers are turning and cooling it. *Robu*
Storage	SanDisk High Endurance 128 GB microSD, 3 off	11,967	Records every frame and detection for a whole mission. Ordinary cards wear out at that rate. *Flipkart*

Item	Model	INR, all 3	Why this part, and from whom
GNSS RTK receiver	Teravolt AeroNav-Pro RTK, aircraft 1 rover + ground base, 2 off	50,000	Centimetre positioning for aircraft 1 and the ground station it corrects against. Only one aircraft carries it: all three are the same machine, so measuring one against surveyed ground is what tells us how well the design works. Quoted by the supplier; this line was an estimate before. *Teravolt*
GNSS receiver	Holybro Micro M9N GPS with case, NEO-M9N and compass, 2 off	13,878	Positioning for aircraft 2 and 3, accurate to about two metres against a five metre requirement. Carries the compass the flight controller needs, and plugs straight into it. *in-house*
Height rangefinder	Benewake TFmini-S, 12 m lidar, UART, 3 off	8,481	Measures true height above whatever is below — water, ground or a rooftop. The kit is released at 6 m, and the barometer measures height above the launch point, which over floodwater is a different thing. *in-house*
Camera + lens	Arducam 12.3 MP 1/2.3 in HQ module, 6 mm CS lens, 3 off	20,397	The search sensor, chosen so a person in water is large enough in the picture to be found. *Robu*
Video transmitter	SunRobotics TS832, 5.8 GHz, 600 mW, 3 off	12,717	Sends the live picture down, one channel per aircraft. *IndiaMART*
Command receiver	ExpressLRS RX24T, 2.4 GHz, 3 off	4,317	Carries the safety pilot's control and the aircraft's reporting on a single link, which saves a second radio. *Robu*
Coordination radio	EByte E22-900T22D-V2, SX1262, 22 dBm, 3 off	2,967	How the three aircraft tell each other what they have found. The most robust link on the aircraft, which is where survivor positions belong. *Robu*
Power module	Holybro PM07, 3 off	15,747	Reports battery voltage and current to the autopilot, which is what triggers a low-battery return. *Robu*
BEC, primary	8 A UBEC, 5/6 V out, 2-6S in, 3 off	11,817	Powers the computer. Deliberately oversized: losing it in flight is the failure this line exists to prevent. *Amazon India*
BEC, secondary	ReadytoSky 2-6S 5V/3A + 12V/3A switchable UBEC, 3 off	837	A separate supply for the servos, so a jammed servo cannot disturb the autopilot. *Robu*

Per aircraft, airframe and drive — phases 7–10, repeated at 17–20 and 25–28 INR 61,186 per aircraft, INR 1,98,224 in all

Item	Model	INR, all 3	Why this part, and from whom
Motors	Tarot TL96020, 5008, 340 KV, 12 off	56,436	Twelve. The first is measured on the stand before the other eleven are bought. *Robokits*
Propellers	1855 counter-rotating carbon fibre, CW+CCW pair, 16 off	26,256	Sixteen for twelve positions. The item most often broken in flight testing, and the whole of our spares policy. *Robokits*
Speed controllers	Darkmatter VISHNU 50 MK-III, 4-in-1, 50 A, 3 off	15,567	One board drives all four motors. Made in India, and previously a quotation. *Robu*
Arm tube	3K carbon fibre, OD25 x ID23 x 1000 mm, 6 off	13,554	The four arms. Carbon, and comfortably strong — the joints, not the tubes, are what limit the airframe. *Robu*
Frame plate stock	3K carbon fibre sheet, 300 x 200 x 2 mm, 3 off	8,685	The arms are tubes; this is the plate they bolt to. One sheet yields the top and bottom of a centre section at this size. *in-house*

Item	Model	INR, all 3	Why this part, and from whom
Motor mounts	Tarot 25 mm motor mount, TL9602, 12 off	13,428	Joins motor to arm without machining the tube. This joint is the part that has to be right. *Robu*
Arm clamps	T-bolt hose clamps 23-25 mm, SS304, 4 pcs, 3 off	9,036	Bolted rather than bonded, so an arm damaged in a heavy landing is swapped in the field instead of scrapping the frame. *Amazon India*
Landing gear	F450/F550 landing skid, 3 off	1,647	Gives clearance underneath for the kit magazine, and a stance wide enough for an untidy landing. *Robu*
Suspension springs	5 mm stainless double torsion spring, 32 off	64	Let the legs absorb a hard landing rather than pass it into the arms and the battery. *IndiaMART*
Printed parts	Pro-Range PETG-CF filament, 1.75 mm, 1 kg	949	Filament for the magazine, mounts and covers. Holds its shape in the sun, which cheaper filament does not. *Robu*
Release servos	MG90S mini servo, 180 deg, 12 off	1,788	Open the four kit stations. Metal gears, because a plastic one can drop a kit without anyone knowing. *Robocraze*
Cells	Molicel INR21700-P45B, 6S3P, 54 off	27,486	The battery: enough for the mission, a second full search, and four minutes in hand. *Robu*
Cell holders	3x1 21700 holder, 21.75 mm bore, 54 off	648	Hold the cells in place and keep them apart, so one failing cell does not take its neighbours with it. *Robu*
Pack interconnect	Pure nickel strip, 0.15 x 27 mm	6,500	Joins the cells. Pure nickel, because the cheaper plated strip runs hot at the current this battery delivers. *IndiaMART*
Group interconnect	Tinned copper braided jumper	500	Joins the battery's groups, where the current is highest. *IndiaMART*
Balance leads	JST-XH 6S, 200 mm, 22 AWG, 3 off	372	How the charger checks each cell. With no battery board fitted, this is our only per-cell check — and it happens on every charge. *zBotic*
Pack fusing	ANL fuse holder + 150 A ANL fuses, 3 off	9,000	Protects against a short circuit without cutting power at full throttle. *Njour*
Pack retention	300 mm LiPo strap, reusable, 6 off	372	Holds the battery down. It is a fifth of the aircraft's weight, so a strap that lets go is not a small problem. *Robu*
Main leads	10 AWG ultra-flexible silicone wire	149	The main power cable, sized for the highest current the motors ever draw rather than the average. *Robu*
Power connectors	Amass XT90S anti-spark, M/F pair, 12 off	2,160	Anti-spark, so connecting a battery this size does not burn the contacts. *Robokits*
Signal connectors	JST ZH 6-pin, 200 mm leads, 3 off	69	Keyed, so the wiring can only go back together one way after a repair. *Robu*
Antenna feeders	SMA M-F bulkhead RG316, 150 mm + adapters, 3 off	3,417	Let the antennas mount on the airframe instead of hanging off the radios. *Desertcart*
Insulation	PVC heat-shrink sleeve, 150 mm, 3 off	141	Sleeving over the battery joints — the one place on the aircraft where a rubbed wire is an immediate fire. *Robu*

Flight-test support — phases 11–12   INR 38,925 of parts

Item	Model	INR, all 3	Why this part, and from whom
Safety-pilot transmitter	RadioMaster TX12 MKII ELRS + RP1 receiver	16,519	Manual override for the safety pilot, independent of the autonomy. A rule requirement and a safety one. *zBotic*

Item	Model	INR, all 3	Why this part, and from whom
Battery charger	ToolKitRC M6D, 500 W, 25 A, dual	7,349	Two batteries recharged between flights is what sets how many tests fit into a day. *Robu*
Pack health monitor	Holybro PM08-CAN, 14S, 200 A	9,058	A bench instrument for tracking how the batteries age over the programme. *Robu*
Cable management	Nylon zip ties, 350 mm, 100 pcs	135	Ties long enough to dress wiring around a one-metre airframe. *Robu*
Heat-shrink kit	328 pcs assorted 2:1 heat-shrink	159	Insulates the signal wiring during assembly. *Robu*
Mounting tape	3M VHB 5952, 6 mm x 8.2 m	560	Holds radios and receivers where a screw would need a fixing the printed part cannot carry. *Amazon India*
Hook and loop	Self-adhesive hook and loop, 25 mm x 2 m	145	For the battery and anything else that must come out between flights without tools. *Amazon India*
Consumables	Fasteners, tape, connectors, filament	5,000	Solder, abrasives, spare hardware, filament. A round figure, because itemising it would be false precision. *in-house*

Ground segment and statutory — phases 29–32   INR 19,436 of parts

Item	Model	INR, all 3	Why this part, and from whom
Video receivers	RC832S 5.8 GHz AV receiver, 3 off	13,197	Three, so the ground station can show all three aircraft at once, which the rules require. *Robu*
Receive antennas	Triple-feed 5.8 GHz patch, SMA, 3 off	2,517	The ground antennas. Most of the video range comes from these rather than from the transmitters. *Robu*
Video capture	USB 2.0 analog AV capture adapter, 3 off	1,197	Brings the three pictures into the ground station computer. *Robu*
Coordination base	EByte E22-900T22U, SX1262, USB	1,646	The ground end of the link the aircraft use to report what they find. *Hubtronics*
Base station mount	Photographic tripod and adapter	879	Holds the ground receiver still during a mission. Survey-grade accuracy is unnecessary; not moving is what matters. *Amazon India*

Why there is no battery management board. An earlier version funded one. Most of what that line covered is already bought under other names, and the part that remains — a board that cuts power when a cell is unhappy — is the wrong device on an aircraft, because it can cut power at full throttle, which is a crash rather than a protection. Instead the battery is fused against short circuits, the autopilot brings the aircraft home on low voltage, and every cell is checked on the charger before each flight. What we do not have is per-cell protection in the air; that is a deliberate choice, not an oversight.

What this arrangement costs us

We should be honest that twenty-nine approvals is a lot of paperwork for the office, and that building one aircraft at a time means ordering the same parts three times over, at single-unit prices and with three lots of shipping. If the paperwork or the price becomes the bigger problem, the per-aircraft steps can be merged into one order.

The detection work needs no money at all: about a week on a departmental GPU and 30 GB of storage, against a dataset that is already public.

7. Future work

Near term. Measure what is currently calculated: detection accuracy, motor thrust, setup time. Evaluate near-infrared sensing — water absorbs strongly and skin does not, so a person appears bright against dark water. A camera without its infrared filter costs INR 3,000, and a public dataset allows evaluation before purchase. Only the training data is flood-specific; the same system suits earthquake, wildfire and avalanche search.

Longer term. A capable aircraft deciding for itself within a small power budget is a hard requirement in planetary aviation. *Ingenuity* flew 72 times on Mars from a 1.8 kg airframe in an atmosphere under one per cent of Earth's density; *Dragonfly* launches in 2028 for Titan. A command takes minutes to arrive, so every decision is made on board. What transfers is what is not flood-specific: onboard survey planning, perception run to an energy budget, and safe behaviour without a link. Microcontroller-class processors, often proposed for such missions, cannot resolve targets of this size, so a planetary configuration needs a different operating point.

8. Conclusion

Three aircraft, one operator, no network: people found in flooded ground and reached with a relief kit, ten hectares in eight minutes, and coordinates a rescue coordinator can act on. The design is finished. What is left is to build it and find out whether it works.

The department is never exposed to more than one step at a time. The first two come to INR 16,996 and settle whether these motors will lift this aircraft. If they will not, we stop there and the department still has a thrust stand. Nothing above INR 26,000 is released until that measurement exists, and the two biggest blocks only come after a complete aircraft has flown.

Floods come back every monsoon, and nothing of this kind exists at a size two people can carry to the water's edge. We would like to begin, and we are asking the department to release Phase 1.

THAPAR AMATEUR ASTRONOMERS SOCIETY

THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY (Deemed to be University)

RescueSwarm — An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

To
The Dean, Student Affairs (DoSA)
Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-1

Date: ____________

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 1 of 29, Establish thrust and mass measurement capability

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 29 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 1 of 29: Establish thrust and mass measurement capability. This is the first step, and is gated only on this approval. We request funding for the following components:

Component	Model	Qty	Cost (INR)
Fasteners	Hex socket-head assortment M3/M4/M5, SS304	1	1,140
Load cell	REES52 20 kg load cell + HX711 breakout	1	854
Threadlocker	Loctite 243, 50 ml	1	664
Thrust-stand mast	SURYAKUSH metal 2.1 m tripod	1	373
Throttle source	Servo tester / ESC consistency tester, PPM	1	111
Load-cell amplifier	7Semi HX711 24-bit ADC	1	104
		Total	**3,246**

We kindly request approval and financial support of **INR 3,246** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
Prabhjot Singh	Robotics and Artificial Intelligence Engg.	2024–2028	1024230021
Manan Kapoor	Computer Science & Engineering	2024–2028	1024030467
Manav Pansari	Electronics & Communication Engineering	2024–2028	1024060151
Ravi Kant Raja	Robotics and Artificial Intelligence Engg.	2024–2028	1024230032
Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

Dr. Surbhi Sharma
Mentor

Dr. Mamata Gulati
Mentor

President, TAAS
Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

For office use — Office of the Dean, Student Affairs

Amount sanctioned (INR)

Signature

Date

THAPAR AMATEUR ASTRONOMERS SOCIETY

THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY (Deemed to be University)

RescueSwarm — An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

To
The Dean, Student Affairs (DoSA)
Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-2

Date: ____________

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 2 of 29, Measure one motor, propeller and speed controller on the stand

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 29 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 2 of 29: Measure one motor, propeller and speed controller on the stand. It falls due only once the previous step has produced its stated result: *instruments commissioned*. We request funding for the following components:

Component	Model	Qty	Cost (INR)
Speed controllers	Darkmatter VISHNU 50 MK-III, 4-in-1, 50 A	1	5,189
Motors	Tarot TL96020, 5008, 340 KV	1	4,703
Propellers	1855 counter-rotating carbon fibre, CW+CCW pair	1	1,641
		Total	**11,533**

We kindly request approval and financial support of **INR 11,533** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
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Manan Kapoor	Computer Science & Engineering	2024–2028	1024030467
Manav Pansari	Electronics & Communication Engineering	2024–2028	1024060151
Ravi Kant Raja	Robotics and Artificial Intelligence Engg.	2024–2028	1024230032
Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

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Thapar Amateur Astronomers Society

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Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

For office use — Office of the Dean, Student Affairs

Amount sanctioned (INR) | Signature | Date

RescueSwarm — An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

To
The Dean, Student Affairs (DoSA)
Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-3

Date: ____________

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 3 of 29, Aircraft 1 – autopilot and control link

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 29 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 3 of 29: Aircraft 1 – autopilot and control link. It falls due only once the previous step has produced its stated result: *thrust verified*. We request funding for the following components:

Component	Model	Qty	Cost (INR)
Flight controller	Holybro Pixhawk 6C Mini	1	22,049
Autopilot log card	SanDisk 32 GB microSDHC, class 10	1	1,549
FC vibration mount	Glass-fibre anti-vibration shock absorber	1	161
		Total	**23,759**

We kindly request approval and financial support of **INR 23,759** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
Prabhjot Singh	Robotics and Artificial Intelligence Engg.	2024–2028	1024230021
Manan Kapoor	Computer Science & Engineering	2024–2028	1024030467
Manav Pansari	Electronics & Communication Engineering	2024–2028	1024060151
Ravi Kant Raja	Robotics and Artificial Intelligence Engg.	2024–2028	1024230032
Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

Dr. Surbhi Sharma
Mentor

Dr. Mamata Gulati
Mentor

President, TAAS
Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

For office use — Office of the Dean, Student Affairs

Amount sanctioned (INR) Signature Date

THAPAR AMATEUR ASTRONOMERS SOCIETY

THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY (Deemed to be University)

RescueSwarm — An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

To
The Dean, Student Affairs (DoSA)
Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-4

Date: ____________

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 4 of 29, Aircraft 1 – onboard computer and AI accelerator

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 29 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 4 of 29: Aircraft 1 – onboard computer and AI accelerator. It falls due only once the previous step has produced its stated result: *autopilot bench-tested*. We request funding for the following components:

Component	Model	Qty	Cost (INR)
Companion computer	Raspberry Pi 5, 8 GB	1	19,999
AI accelerator	Raspberry Pi AI HAT+, 26 TOPS	1	11,749
Storage	SanDisk High Endurance 128 GB microSD	1	3,989
Compute cooling	Official Raspberry Pi 5 active cooler	1	519
		Total	**36,256**

We kindly request approval and financial support of **INR 36,256** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
Prabhjot Singh	Robotics and Artificial Intelligence Engg.	2024–2028	1024230021
Manan Kapoor	Computer Science & Engineering	2024–2028	1024030467
Manav Pansari	Electronics & Communication Engineering	2024–2028	1024060151
Ravi Kant Raja	Robotics and Artificial Intelligence Engg.	2024–2028	1024230032
Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

Dr. Surbhi Sharma
Mentor

Dr. Mamata Gulati
Mentor

President, TAAS
Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

For office use — Office of the Dean, Student Affairs

Amount sanctioned (INR)

Signature

Date

RescueSwarm — An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

To
The Dean, Student Affairs (DoSA)
Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-5

Date: ____________

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 5 of 29, Aircraft 1 – centimetre positioning: RTK rover and ground base

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 29 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 5 of 29: Aircraft 1 – centimetre positioning: RTK rover and ground base. It falls due only once the previous step has produced its stated result: *compute stack operating*. We request funding for the following components:

Component	Model	Qty	Cost (INR)
GNSS RTK receiver	Teravolt AeroNav-Pro RTK, aircraft 1 rover + ground base	2	50,000
		Total	**50,000**

We kindly request approval and financial support of **INR 50,000** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
Prabhjot Singh	Robotics and Artificial Intelligence Engg.	2024–2028	1024230021
Manan Kapoor	Computer Science & Engineering	2024–2028	1024030467
Manav Pansari	Electronics & Communication Engineering	2024–2028	1024060151
Ravi Kant Raja	Robotics and Artificial Intelligence Engg.	2024–2028	1024230032
Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

Dr. Surbhi Sharma
Mentor

Dr. Mamata Gulati
Mentor

President, TAAS
Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

For office use — Office of the Dean, Student Affairs

Amount sanctioned (INR)

Signature

Date

THAPAR AMATEUR ASTRONOMERS SOCIETY

THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY (Deemed to be University)

RescueSwarm — An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

To
The Dean, Student Affairs (DoSA)
Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-6

Date: ________________

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 6 of 29, Aircraft 1 – camera and the three radios

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 29 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 6 of 29: Aircraft 1 – camera and the three radios. It falls due only once the previous step has produced its stated result: *position fix acquired*. We request funding for the following components:

Component	Model	Qty	Cost (INR)
Camera + lens	Arducam 12.3 MP 1/2.3 in HQ module, 6 mm CS lens	1	6,799
Video transmitter	SunRobotics TS832, 5.8 GHz, 600 mW	1	4,239
Height rangefinder	Benewake TFmini-S, 12 m lidar, UART	1	2,827
Command receiver	ExpressLRS RX24T, 2.4 GHz	1	1,439
Coordination radio	EByte E22-900T22D-V2, SX1262, 22 dBm	1	989
		Total	**16,293**

We kindly request approval and financial support of **INR 16,293** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
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Manav Pansari	Electronics & Communication Engineering	2024–2028	1024060151
Ravi Kant Raja	Robotics and Artificial Intelligence Engg.	2024–2028	1024230032
Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

Dr. Surbhi Sharma
Mentor

Dr. Mamata Gulati
Mentor

President, TAAS
Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

For office use — Office of the Dean, Student Affairs

Amount sanctioned (INR)

Signature

Date

RescueSwarm — An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

To
The Dean, Student Affairs (DoSA)
Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-7

Date: ____________

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 7 of 29, Aircraft 1 – airframe fabrication

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 29 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 7 of 29: Aircraft 1 – airframe fabrication. It falls due only once the previous step has produced its stated result: *avionics integrated*. We request funding for the following components:

Component	Model	Qty	Cost (INR)
Arm tube	3K carbon fibre, OD25 x ID23 x 1000 mm	2	4,518
Motor mounts	Tarot 25 mm motor mount, TL9602	4	4,476
Arm clamps	T-bolt hose clamps 23-25 mm, SS304, 4 pcs	1	3,012
Frame plate stock	3K carbon fibre sheet, 300 x 200 x 2 mm	1	2,895
Printed parts	Pro-Range PETG-CF filament, 1.75 mm, 1 kg	1	949
Landing gear	F450/F550 landing skid	1	549
Suspension springs	5 mm stainless double torsion spring	10	20
		Total	**16,419**

We kindly request approval and financial support of **INR 16,419** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
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Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

Dr. Surbhi Sharma
Mentor

Dr. Mamata Gulati
Mentor

President, TAAS
Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

For office use — Office of the Dean, Student Affairs

Amount sanctioned (INR)

Signature

Date

RescueSwarm — An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

To
The Dean, Student Affairs (DoSA)
Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-8
Date: ____________

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 8 of 29, Aircraft 1 – battery pack and power distribution

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 29 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 8 of 29: Aircraft 1 – battery pack and power distribution. It falls due only once the previous step has produced its stated result: *airframe fabricated*. We request funding for the following components:

Component	Model	Qty	Cost (INR)
Cells	Molicel INR21700-P45B, 6S3P	18	9,162
Pack interconnect	Pure nickel strip, 0.15 x 27 mm	1	6,500
Power module	Holybro PM07	1	5,249
BEC, primary	8 A UBEC, 5/6 V out, 2-6S in	1	3,939
Pack fusing	ANL fuse holder + 150 A ANL fuses	1	3,000
Group interconnect	Tinned copper braided jumper	1	500
BEC, secondary	ReadytoSky 2-6S 5V/3A + 12V/3A switchable UBEC	1	279
Cell holders	3x1 21700 holder, 21.75 mm bore	18	216
Pack retention	300 mm LiPo strap, reusable	2	124
Balance leads	JST-XH 6S, 200 mm, 22 AWG	1	124
		Total	**29,093**

We kindly request approval and financial support of **INR 29,093** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
Prabhjot Singh	Robotics and Artificial Intelligence Engg.	2024–2028	1024230021
Manan Kapoor	Computer Science & Engineering	2024–2028	1024030467
Manav Pansari	Electronics & Communication Engineering	2024–2028	1024060151
Ravi Kant Raja	Robotics and Artificial Intelligence Engg.	2024–2028	1024230032
Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

Dr. Surbhi Sharma
Mentor

Dr. Mamata Gulati
Mentor

President, TAAS
Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

For office use — Office of the Dean, Student Affairs

Amount sanctioned (INR)

Signature

Date

THAPAR AMATEUR ASTRONOMERS SOCIETY

THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY (Deemed to be University)

RescueSwarm — An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

To
The Dean, Student Affairs (DoSA)
Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-9

Date: ____________

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 9 of 29, Aircraft 1 – release servos and wiring

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 29 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 9 of 29: Aircraft 1 – release servos and wiring. It falls due only once the previous step has produced its stated result: *pack bench-discharged*. We request funding for the following components:

Component	Model	Qty	Cost (INR)
Antenna feeders	SMA M-F bulkhead RG316, 150 mm + adapters	1	1,139
Power connectors	Amass XT90S anti-spark, M/F pair	4	720
Release servos	MG90S mini servo, 180 deg	4	596
Main leads	10 AWG ultra-flexible silicone wire	1	149
Insulation	PVC heat-shrink sleeve, 150 mm	1	47
Signal connectors	JST ZH 6-pin, 200 mm leads	1	23
		Total	**2,674**

We kindly request approval and financial support of **INR 2,674** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
Prabhjot Singh	Robotics and Artificial Intelligence Engg.	2024–2028	1024230021
Manan Kapoor	Computer Science & Engineering	2024–2028	1024030467
Manav Pansari	Electronics & Communication Engineering	2024–2028	1024060151
Ravi Kant Raja	Robotics and Artificial Intelligence Engg.	2024–2028	1024230032
Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

Dr. Surbhi Sharma
Mentor

Dr. Mamata Gulati
Mentor

President, TAAS
Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

For office use — Office of the Dean, Student Affairs

Amount sanctioned (INR)

Signature

Date

THAPAR AMATEUR ASTRONOMERS SOCIETY

THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY (Deemed to be University)

RescueSwarm — An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

To
The Dean, Student Affairs (DoSA)
Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-10

Date: ____________________

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 10 of 29, Aircraft 1 – propulsion completed

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 29 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 10 of 29: Aircraft 1 – propulsion completed. It falls due only once the previous step has produced its stated result: *drive train installed*. We request funding for the following components:

Component	Model	Qty	Cost (INR)
Motors	Tarot TL96020, 5008, 340 KV	3	14,109
Propellers	1855 counter-rotating carbon fibre, CW+CCW pair	5	8,205
		Total	**22,314**

We kindly request approval and financial support of **INR 22,314** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
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Manav Pansari	Electronics & Communication Engineering	2024–2028	1024060151
Ravi Kant Raja	Robotics and Artificial Intelligence Engg.	2024–2028	1024230032
Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

Dr. Surbhi Sharma
Mentor

Dr. Mamata Gulati
Mentor

President, TAAS
Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

For office use — Office of the Dean, Student Affairs

Amount sanctioned (INR)

Signature

Date

RescueSwarm — An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

To
The Dean, Student Affairs (DoSA)
Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-11

Date: ____________________

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 11 of 29, Charging, pack instrumentation and manual override

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 29 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 11 of 29: Charging, pack instrumentation and manual override. It falls due only once the previous step has produced its stated result: *aircraft assembled and weighed.* We request funding for the following components:

Component	Model	Qty	Cost (INR)
Safety-pilot transmitter	RadioMaster TX12 MKII ELRS + RP1 receiver	1	16,519
Pack health monitor	Holybro PM08-CAN, 14S, 200 A	1	9,058
Battery charger	ToolKitRC M6D, 500 W, 25 A, dual	1	7,349
		Total	**32,926**

We kindly request approval and financial support of **INR 32,926** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
Prabhjot Singh	Robotics and Artificial Intelligence Engg.	2024–2028	1024230021
Manan Kapoor	Computer Science & Engineering	2024–2028	1024030467
Manav Pansari	Electronics & Communication Engineering	2024–2028	1024060151
Ravi Kant Raja	Robotics and Artificial Intelligence Engg.	2024–2028	1024230032
Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

Dr. Surbhi Sharma
Mentor

Dr. Mamata Gulati
Mentor

President, TAAS
Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

For office use — Office of the Dean, Student Affairs

Amount sanctioned (INR)

Signature

Date

THAPAR AMATEUR ASTRONOMERS SOCIETY

THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY (Deemed to be University)

RescueSwarm — An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

To
The Dean, Student Affairs (DoSA)
Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-12

Date: ________________

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 12 of 29, Field consumables and assembly materials

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 29 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 12 of 29: Field consumables and assembly materials. It falls due only once the previous step has produced its stated result: *charging verified*. We request funding for the following components:

Component	Model	Qty	Cost (INR)
Consumables	Fasteners, tape, connectors, filament	1	5,000
Mounting tape	3M VHB 5952, 6 mm x 8.2 m	1	560
Heat-shrink kit	328 pcs assorted 2:1 heat-shrink	1	159
Hook and loop	Self-adhesive hook and loop, 25 mm x 2 m	1	145
Cable management	Nylon zip ties, 350 mm, 100 pcs	1	135
		Total	**5,999**

We kindly request approval and financial support of **INR 5,999** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
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Manan Kapoor	Computer Science & Engineering	2024–2028	1024030467
Manav Pansari	Electronics & Communication Engineering	2024–2028	1024060151
Ravi Kant Raja	Robotics and Artificial Intelligence Engg.	2024–2028	1024230032
Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

Dr. Surbhi Sharma
Mentor

Dr. Mamata Gulati
Mentor

President, TAAS
Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

For office use — Office of the Dean, Student Affairs

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THAPAR AMATEUR ASTRONOMERS SOCIETY

THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY (Deemed to be University)

RescueSwarm — An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

To
The Dean, Student Affairs (DoSA)
Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-13

Date: ____________

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 13 of 29, Aircraft 2 – autopilot and control link

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 29 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 13 of 29: Aircraft 2 – autopilot and control link. It falls due only once the previous step has produced its stated result: *aircraft 1 flew a full mission*. We request funding for the following components:

Component	Model	Qty	Cost (INR)
Flight controller	Holybro Pixhawk 6C Mini	1	22,049
Autopilot log card	SanDisk 32 GB microSDHC, class 10	1	1,549
FC vibration mount	Glass-fibre anti-vibration shock absorber	1	161
		Total	**23,759**

We kindly request approval and financial support of **INR 23,759** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
Prabhjot Singh	Robotics and Artificial Intelligence Engg.	2024–2028	1024230021
Manan Kapoor	Computer Science & Engineering	2024–2028	1024030467
Manav Pansari	Electronics & Communication Engineering	2024–2028	1024060151
Ravi Kant Raja	Robotics and Artificial Intelligence Engg.	2024–2028	1024230032
Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

Dr. Surbhi Sharma
Mentor

Dr. Mamata Gulati
Mentor

President, TAAS
Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

For office use — Office of the Dean, Student Affairs

Amount sanctioned (INR)

Signature

Date

RescueSwarm — An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

To
The Dean, Student Affairs (DoSA)
Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-14

Date: ____________

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 14 of 29, Aircraft 2 – onboard computer and AI accelerator

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 29 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 14 of 29: Aircraft 2 – onboard computer and AI accelerator. It falls due only once the previous step has produced its stated result: *autopilot bench-tested*. We request funding for the following components:

Component	Model	Qty	Cost (INR)
Companion computer	Raspberry Pi 5, 8 GB	1	19,999
AI accelerator	Raspberry Pi AI HAT+, 26 TOPS	1	11,749
Storage	SanDisk High Endurance 128 GB microSD	1	3,989
Compute cooling	Official Raspberry Pi 5 active cooler	1	519
		Total	**36,256**

We kindly request approval and financial support of **INR 36,256** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
Prabhjot Singh	Robotics and Artificial Intelligence Engg.	2024–2028	1024230021
Manan Kapoor	Computer Science & Engineering	2024–2028	1024030467
Manav Pansari	Electronics & Communication Engineering	2024–2028	1024060151
Ravi Kant Raja	Robotics and Artificial Intelligence Engg.	2024–2028	1024230032
Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

Dr. Surbhi Sharma
Mentor

Dr. Mamata Gulati
Mentor

President, TAAS
Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

For office use — Office of the Dean, Student Affairs

Amount sanctioned (INR)

Signature

Date

THAPAR AMATEUR ASTRONOMERS SOCIETY

THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY (Deemed to be University)

RescueSwarm — An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

To
The Dean, Student Affairs (DoSA)
Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-15

Date: ____________

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 15 of 29, Aircraft 2 – satellite positioning and compass

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 29 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 15 of 29: Aircraft 2 – satellite positioning and compass. It falls due only once the previous step has produced its stated result: *compute stack operating*. We request funding for the following components:

Component	Model	Qty	Cost (INR)
GNSS receiver	Holybro Micro M9N GPS with case, NEO-M9N and compass	1	6,939
		Total	**6,939**

We kindly request approval and financial support of **INR 6,939** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
Prabhjot Singh	Robotics and Artificial Intelligence Engg.	2024–2028	1024230021
Manan Kapoor	Computer Science & Engineering	2024–2028	1024030467
Manav Pansari	Electronics & Communication Engineering	2024–2028	1024060151
Ravi Kant Raja	Robotics and Artificial Intelligence Engg.	2024–2028	1024230032
Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

Dr. Surbhi Sharma
Mentor

Dr. Mamata Gulati
Mentor

President, TAAS
Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

For office use — Office of the Dean, Student Affairs

Amount sanctioned (INR)

Signature

Date

THAPAR AMATEUR ASTRONOMERS SOCIETY

THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY (Deemed to be University)

RescueSwarm — An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

To
The Dean, Student Affairs (DoSA)
Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-16

Date: ____________

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 16 of 29, Aircraft 2 – camera and the three radios

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 29 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 16 of 29: Aircraft 2 – camera and the three radios. It falls due only once the previous step has produced its stated result: *position fix acquired*. We request funding for the following components:

Component	Model	Qty	Cost (INR)
Camera + lens	Arducam 12.3 MP 1/2.3 in HQ module, 6 mm CS lens	1	6,799
Video transmitter	SunRobotics TS832, 5.8 GHz, 600 mW	1	4,239
Height rangefinder	Benewake TFmini-S, 12 m lidar, UART	1	2,827
Command receiver	ExpressLRS RX24T, 2.4 GHz	1	1,439
Coordination radio	EByte E22-900T22D-V2, SX1262, 22 dBm	1	989
		Total	**16,293**

We kindly request approval and financial support of **INR 16,293** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
Prabhjot Singh	Robotics and Artificial Intelligence Engg.	2024–2028	1024230021
Manan Kapoor	Computer Science & Engineering	2024–2028	1024030467
Manav Pansari	Electronics & Communication Engineering	2024–2028	1024060151
Ravi Kant Raja	Robotics and Artificial Intelligence Engg.	2024–2028	1024230032
Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

Dr. Surbhi Sharma
Mentor

Dr. Mamata Gulati
Mentor

President, TAAS
Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

For office use — Office of the Dean, Student Affairs

Amount sanctioned (INR)

Signature

Date

THAPAR AMATEUR ASTRONOMERS SOCIETY

THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY (Deemed to be University)

RescueSwarm — An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

To
The Dean, Student Affairs (DoSA)
Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-17

Date: ____________

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 17 of 29, Aircraft 2 – airframe fabrication

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 29 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 17 of 29: Aircraft 2 – airframe fabrication. It falls due only once the previous step has produced its stated result: *avionics integrated*. We request funding for the following components:

Component	Model	Qty	Cost (INR)
Arm tube	3K carbon fibre, OD25 x ID23 x 1000 mm	2	4,518
Motor mounts	Tarot 25 mm motor mount, TL9602	4	4,476
Arm clamps	T-bolt hose clamps 23-25 mm, SS304, 4 pcs	1	3,012
Frame plate stock	3K carbon fibre sheet, 300 x 200 x 2 mm	1	2,895
Landing gear	F450/F550 landing skid	1	549
Suspension springs	5 mm stainless double torsion spring	10	20
		Total	**15,470**

We kindly request approval and financial support of **INR 15,470** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
Prabhjot Singh	Robotics and Artificial Intelligence Engg.	2024–2028	1024230021
Manan Kapoor	Computer Science & Engineering	2024–2028	1024030467
Manav Pansari	Electronics & Communication Engineering	2024–2028	1024060151
Ravi Kant Raja	Robotics and Artificial Intelligence Engg.	2024–2028	1024230032
Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

Dr. Surbhi Sharma
Mentor

Dr. Mamata Gulati
Mentor

President, TAAS
Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

For office use — Office of the Dean, Student Affairs

Amount sanctioned (INR)

Signature

Date

THAPAR AMATEUR ASTRONOMERS SOCIETY

THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY (Deemed to be University)

RescueSwarm — An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

To
The Dean, Student Affairs (DoSA)
Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-18

Date: ____________

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 18 of 29, Aircraft 2 – battery pack and power distribution

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 29 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 18 of 29: Aircraft 2 – battery pack and power distribution. It falls due only once the previous step has produced its stated result: *airframe fabricated*. We request funding for the following components:

Component	Model	Qty	Cost (INR)
Cells	Molicel INR21700-P45B, 6S3P	18	9,162
Power module	Holybro PM07	1	5,249
BEC, primary	8 A UBEC, 5/6 V out, 2-6S in	1	3,939
Pack fusing	ANL fuse holder + 150 A ANL fuses	1	3,000
BEC, secondary	ReadytoSky 2-6S 5V/3A + 12V/3A switchable UBEC	1	279
Cell holders	3x1 21700 holder, 21.75 mm bore	18	216
Pack retention	300 mm LiPo strap, reusable	2	124
Balance leads	JST-XH 6S, 200 mm, 22 AWG	1	124
		Total	**22,093**

We kindly request approval and financial support of **INR 22,093** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
Prabhjot Singh	Robotics and Artificial Intelligence Engg.	2024–2028	1024230021
Manan Kapoor	Computer Science & Engineering	2024–2028	1024030467
Manav Pansari	Electronics & Communication Engineering	2024–2028	1024060151
Ravi Kant Raja	Robotics and Artificial Intelligence Engg.	2024–2028	1024230032
Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

Dr. Surbhi Sharma
Mentor

Dr. Mamata Gulati
Mentor

President, TAAS
Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

For office use — Office of the Dean, Student Affairs

Amount sanctioned (INR)

Signature

Date

THAPAR AMATEUR ASTRONOMERS SOCIETY

THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY (Deemed to be University)

RescueSwarm — An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

To
The Dean, Student Affairs (DoSA)
Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-19

Date: ____________

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 19 of 29, Aircraft 2 – speed controllers, release servos, wiring

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 29 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 19 of 29: Aircraft 2 – speed controllers, release servos, wiring. It falls due only once the previous step has produced its stated result: *pack bench-discharged.* We request funding for the following components:

Component	Model	Qty	Cost (INR)
Speed controllers	Darkmatter VISHNU 50 MK-III, 4-in-1, 50 A	1	5,189
Antenna feeders	SMA M-F bulkhead RG316, 150 mm + adapters	1	1,139
Power connectors	Amass XT90S anti-spark, M/F pair	4	720
Release servos	MG90S mini servo, 180 deg	4	596
Insulation	PVC heat-shrink sleeve, 150 mm	1	47
Signal connectors	JST ZH 6-pin, 200 mm leads	1	23
		Total	**7,714**

We kindly request approval and financial support of **INR 7,714** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
Prabhjot Singh	Robotics and Artificial Intelligence Engg.	2024–2028	1024230021
Manan Kapoor	Computer Science & Engineering	2024–2028	1024030467
Manav Pansari	Electronics & Communication Engineering	2024–2028	1024060151
Ravi Kant Raja	Robotics and Artificial Intelligence Engg.	2024–2028	1024230032
Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

Dr. Surbhi Sharma
Mentor

Dr. Mamata Gulati
Mentor

President, TAAS
Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

For office use — Office of the Dean, Student Affairs

Amount sanctioned (INR) | Signature | Date

THAPAR AMATEUR ASTRONOMERS SOCIETY

THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY (Deemed to be University)

RescueSwarm — An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

To
The Dean, Student Affairs (DoSA)
Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-20

Date: ____________

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 20 of 29, Aircraft 2 – propulsion completed

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 29 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 20 of 29: Aircraft 2 – propulsion completed. It falls due only once the previous step has produced its stated result: *drive train installed*. We request funding for the following components:

Component	Model	Qty	Cost (INR)
Motors	Tarot TL96020, 5008, 340 KV	4	18,812
Propellers	1855 counter-rotating carbon fibre, CW+CCW pair	5	8,205
		Total	**27,017**

We kindly request approval and financial support of **INR 27,017** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
Prabhjot Singh	Robotics and Artificial Intelligence Engg.	2024–2028	1024230021
Manan Kapoor	Computer Science & Engineering	2024–2028	1024030467
Manav Pansari	Electronics & Communication Engineering	2024–2028	1024060151
Ravi Kant Raja	Robotics and Artificial Intelligence Engg.	2024–2028	1024230032
Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

Dr. Surbhi Sharma
Mentor

Dr. Mamata Gulati
Mentor

President, TAAS
Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

For office use — Office of the Dean, Student Affairs

Amount sanctioned (INR)

Signature

Date

THAPAR AMATEUR ASTRONOMERS SOCIETY

THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY (Deemed to be University)

RescueSwarm — An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

To
The Dean, Student Affairs (DoSA)
Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-21

Date: ____________

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 21 of 29, Aircraft 3 – autopilot and control link

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 29 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 21 of 29: Aircraft 3 – autopilot and control link. It falls due only once the previous step has produced its stated result: *two aircraft, separation verified*. We request funding for the following components:

Component	Model	Qty	Cost (INR)
Flight controller	Holybro Pixhawk 6C Mini	1	22,049
Autopilot log card	SanDisk 32 GB microSDHC, class 10	1	1,549
FC vibration mount	Glass-fibre anti-vibration shock absorber	1	161
		Total	**23,759**

We kindly request approval and financial support of **INR 23,759** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
Prabhjot Singh	Robotics and Artificial Intelligence Engg.	2024–2028	1024230021
Manan Kapoor	Computer Science & Engineering	2024–2028	1024030467
Manav Pansari	Electronics & Communication Engineering	2024–2028	1024060151
Ravi Kant Raja	Robotics and Artificial Intelligence Engg.	2024–2028	1024230032
Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

Dr. Surbhi Sharma
Mentor

Dr. Mamata Gulati
Mentor

President, TAAS
Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

For office use — Office of the Dean, Student Affairs

Amount sanctioned (INR)

Signature

Date

RescueSwarm — An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

To
The Dean, Student Affairs (DoSA)
Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-22

Date: ____________

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 22 of 29, Aircraft 3 – onboard computer and AI accelerator

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 29 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 22 of 29: Aircraft 3 – onboard computer and AI accelerator. It falls due only once the previous step has produced its stated result: *autopilot bench-tested*. We request funding for the following components:

Component	Model	Qty	Cost (INR)
Companion computer	Raspberry Pi 5, 8 GB	1	19,999
AI accelerator	Raspberry Pi AI HAT+, 26 TOPS	1	11,749
Storage	SanDisk High Endurance 128 GB microSD	1	3,989
Compute cooling	Official Raspberry Pi 5 active cooler	1	519
		Total	**36,256**

We kindly request approval and financial support of **INR 36,256** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
Prabhjot Singh	Robotics and Artificial Intelligence Engg.	2024–2028	1024230021
Manan Kapoor	Computer Science & Engineering	2024–2028	1024030467
Manav Pansari	Electronics & Communication Engineering	2024–2028	1024060151
Ravi Kant Raja	Robotics and Artificial Intelligence Engg.	2024–2028	1024230032
Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

Dr. Surbhi Sharma
Mentor

Dr. Mamata Gulati
Mentor

President, TAAS
Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

For office use — Office of the Dean, Student Affairs

Amount sanctioned (INR)

Signature

Date

RescueSwarm — An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

To
The Dean, Student Affairs (DoSA)
Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-23

Date: ____________

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 23 of 29, Aircraft 3 – satellite positioning and compass

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 29 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 23 of 29: Aircraft 3 – satellite positioning and compass. It falls due only once the previous step has produced its stated result: *compute stack operating*. We request funding for the following components:

Component	Model	Qty	Cost (INR)
GNSS receiver	Holybro Micro M9N GPS with case, NEO-M9N and compass	1	6,939
		Total	**6,939**

We kindly request approval and financial support of **INR 6,939** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
Prabhjot Singh	Robotics and Artificial Intelligence Engg.	2024–2028	1024230021
Manan Kapoor	Computer Science & Engineering	2024–2028	1024030467
Manav Pansari	Electronics & Communication Engineering	2024–2028	1024060151
Ravi Kant Raja	Robotics and Artificial Intelligence Engg.	2024–2028	1024230032
Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

Dr. Surbhi Sharma
Mentor

Dr. Mamata Gulati
Mentor

President, TAAS
Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

For office use — Office of the Dean, Student Affairs

Amount sanctioned (INR)

Signature

Date

THAPAR AMATEUR ASTRONOMERS SOCIETY

THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY (Deemed to be University)

RescueSwarm — An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

To
The Dean, Student Affairs (DoSA)
Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-24

Date: ____________

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 24 of 29, Aircraft 3 – camera and the three radios

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 29 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 24 of 29: Aircraft 3 – camera and the three radios. It falls due only once the previous step has produced its stated result: *position fix acquired*. We request funding for the following components:

Component	Model	Qty	Cost (INR)
Camera + lens	Arducam 12.3 MP 1/2.3 in HQ module, 6 mm CS lens	1	6,799
Video transmitter	SunRobotics TS832, 5.8 GHz, 600 mW	1	4,239
Height rangefinder	Benewake TFmini-S, 12 m lidar, UART	1	2,827
Command receiver	ExpressLRS RX24T, 2.4 GHz	1	1,439
Coordination radio	EByte E22-900T22D-V2, SX1262, 22 dBm	1	989
		Total	**16,293**

We kindly request approval and financial support of **INR 16,293** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
Prabhjot Singh	Robotics and Artificial Intelligence Engg.	2024–2028	1024230021
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Manav Pansari	Electronics & Communication Engineering	2024–2028	1024060151
Ravi Kant Raja	Robotics and Artificial Intelligence Engg.	2024–2028	1024230032
Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

Dr. Surbhi Sharma
Mentor

Dr. Mamata Gulati
Mentor

President, TAAS
Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

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Date

RescueSwarm — An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

To
The Dean, Student Affairs (DoSA)
Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-25

Date: ______________

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 25 of 29, Aircraft 3 – airframe fabrication

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 29 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 25 of 29: Aircraft 3 – airframe fabrication. It falls due only once the previous step has produced its stated result: *avionics integrated.* We request funding for the following components:

Component	Model	Qty	Cost (INR)
Arm tube	3K carbon fibre, OD25 x ID23 x 1000 mm	2	4,518
Motor mounts	Tarot 25 mm motor mount, TL9602	4	4,476
Arm clamps	T-bolt hose clamps 23-25 mm, SS304, 4 pcs	1	3,012
Frame plate stock	3K carbon fibre sheet, 300 x 200 x 2 mm	1	2,895
Landing gear	F450/F550 landing skid	1	549
Suspension springs	5 mm stainless double torsion spring	12	24
		Total	**15,474**

We kindly request approval and financial support of **INR 15,474** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

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Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
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Manan Kapoor	Computer Science & Engineering	2024–2028	1024030467
Manav Pansari	Electronics & Communication Engineering	2024–2028	1024060151
Ravi Kant Raja	Robotics and Artificial Intelligence Engg.	2024–2028	1024230032
Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

Dr. Surbhi Sharma
Mentor

Dr. Mamata Gulati
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President, TAAS
Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

For office use — Office of the Dean, Student Affairs

Amount sanctioned (INR)

Signature

Date

THAPAR AMATEUR ASTRONOMERS SOCIETY

THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY (Deemed to be University)

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To
The Dean, Student Affairs (DoSA)
Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-26

Date: ____________

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 26 of 29, Aircraft 3 – battery pack and power distribution

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 29 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 26 of 29: Aircraft 3 – battery pack and power distribution. It falls due only once the previous step has produced its stated result: *airframe fabricated*. We request funding for the following components:

Component	Model	Qty	Cost (INR)
Cells	Molicel INR21700-P45B, 6S3P	18	9,162
Power module	Holybro PM07	1	5,249
BEC, primary	8 A UBEC, 5/6 V out, 2-6S in	1	3,939
Pack fusing	ANL fuse holder + 150 A ANL fuses	1	3,000
BEC, secondary	ReadytoSky 2-6S 5V/3A + 12V/3A switchable UBEC	1	279
Cell holders	3x1 21700 holder, 21.75 mm bore	18	216
Pack retention	300 mm LiPo strap, reusable	2	124
Balance leads	JST-XH 6S, 200 mm, 22 AWG	1	124
		Total	**22,093**

We kindly request approval and financial support of **INR 22,093** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
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Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

Dr. Surbhi Sharma
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Dr. Mamata Gulati
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President, TAAS
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Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

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THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY (Deemed to be University)

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To
The Dean, Student Affairs (DoSA)
Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-27

Date: ____________

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 27 of 29, Aircraft 3 – speed controllers, release servos, wiring

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 29 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 27 of 29: Aircraft 3 – speed controllers, release servos, wiring. It falls due only once the previous step has produced its stated result: *pack bench-discharged.* We request funding for the following components:

Component	Model	Qty	Cost (INR)
Speed controllers	Darkmatter VISHNU 50 MK-III, 4-in-1, 50 A	1	5,189
Antenna feeders	SMA M-F bulkhead RG316, 150 mm + adapters	1	1,139
Power connectors	Amass XT90S anti-spark, M/F pair	4	720
Release servos	MG90S mini servo, 180 deg	4	596
Insulation	PVC heat-shrink sleeve, 150 mm	1	47
Signal connectors	JST ZH 6-pin, 200 mm leads	1	23
		Total	**7,714**

We kindly request approval and financial support of **INR 7,714** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
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Approvals & Signatures

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Thapar Amateur Astronomers Society

Swastik Kumar
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Submitted on behalf of the team

Date

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To
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Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-28

Date: ____________________

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 28 of 29, Aircraft 3 – propulsion completed

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 29 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 28 of 29: Aircraft 3 – propulsion completed. It falls due only once the previous step has produced its stated result: *drive train installed*. We request funding for the following components:

Component	Model	Qty	Cost (INR)
Motors	Tarot TL96020, 5008, 340 KV	4	18,812
Propellers	1855 counter-rotating carbon fibre, CW+CCW pair	5	8,205
		Total	**27,017**

We kindly request approval and financial support of **INR 27,017** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

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Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
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Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

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Dr. Mamata Gulati
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Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

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Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-29

Date: ________________

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 29 of 29, Ground station: three video feeds, data link, base mount

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 29 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 29 of 29: Ground station: three video feeds, data link, base mount. It falls due only once the previous step has produced its stated result: *third aircraft flying*. We request funding for the following components:

Component	Model	Qty	Cost (INR)
Video receivers	RC832S 5.8 GHz AV receiver	3	13,197
Receive antennas	Triple-feed 5.8 GHz patch, SMA	3	2,517
Coordination base	EByte E22-900T22U, SX1262, USB	1	1,646
Video capture	USB 2.0 analog AV capture adapter	3	1,197
Base station mount	Photographic tripod and adapter	1	879
		Total	**19,436**

We kindly request approval and financial support of **INR 19,436** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

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Manan Kapoor	Computer Science & Engineering	2024–2028	1024030467
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